

Tumorigenesis and Tumor Progression

Five stages of malignant transformation and tumor progression in melanocytes have been suggested, based on clinical, histopathologic, immunopathologic, cytogenetic, and in vitro properties:

- (1) benign melanocytic nevi;
- (2) atypical nevi;
- (3) primary malignant melanoma, radial growth phase;
- (4) primary malignant melanoma, vertical growth phase; and
- (5) metastatic malignant melanoma.

It is believed that with each successive step of tumorigenesis, a new clone of cells emerges with growth advantages over the surrounding tissue, resulting in "clonal expansion." A critical step in melanoma progression may be the transition from the radial to the vertical growth phase.

The **radial growth phase** involves primarily intraepidermal proliferation of melanoma cells, along with limited invasion into the papillary dermis by a small number of cells that have acquired a growth advantage. These cells can proliferate autonomously in this location but do not form large aggregates. Radial growth phase cells express **E-cadherin**, an adhesion molecule that mediates interactions with keratinocytes and restricts cell migration from the epidermis. Melanomas at this stage are generally considered incapable of metastasis.

The **vertical growth phase** is defined by **aggregative growth**, leading to the formation of expansile nests or nodules of tumor cells. This phase is associated with increased tumorigenic potential and correlates more closely with metastatic capacity, although it is likely an oversimplification of the full metastatic cascade. A key molecular change in this phase is the loss of E-cadherin and the gain of **N-cadherin**, which enables interactions with

fibroblasts, macrophages, and endothelial cells. This switch in cadherin expression may promote stromal invasion and intravasation of malignant cells.

Understanding melanoma tumorigenesis requires consideration of several clinical observations:

- (1) Precursor nevi are associated with melanoma in approximately 30% of cases, while 70% arise without a visible precursor;
- (2) UV radiation plays a significant role in pathogenesis;
- (3) Patient pigmentary phenotype (e.g., fair skin, tendency to sunburn) influences risk; and
- (4) Family history and genetic factors contribute to susceptibility.

These observations support at least two major pathways of melanomagenesis.

In the **first pathway**, melanomas—particularly **superficial spreading melanomas (SSMs)**—arise in association with pre-existing melanocytic nevi. Nevi may represent an early clonal proliferation or an initiated lesion. UV exposure, especially in childhood, is a likely initiating event. Epidemiologic data suggest early-life sun exposure increases nevus count, thereby expanding the pool of cells at risk for subsequent mutations. As UV radiation acts as a complete carcinogen, intermittent sun exposure may drive additional mutational events, leading to malignant transformation.

The **second pathway** is exemplified by **lentigo maligna melanoma (LMM)**, which develops on chronically sun-exposed skin due to cumulative UV damage. This type shows a steady increase in incidence with age, consistent with prolonged, repeated DNA injury in melanocytes. The pattern supports a model of continuous initiation by UV radiation.

Curtin et al. analyzed genome-wide copy number alterations and mutational status of **BRAF** and **NRAS** in melanomas from patients with varying levels of sun exposure. Their findings revealed distinct genetic profiles based on anatomical site and UV exposure, indicating multiple genetic pathways in melanoma development. This supports the concept that clinical heterogeneity in melanoma reflects genetically distinct subtypes with differing sensitivities to UV radiation. Additional, yet unidentified, UV-independent pathways may also exist.

Clinical Findings

Sub-Types of Melanoma

Superficial Spreading Melanoma

Superficial spreading melanoma (SSM) is the most common subtype, accounting for approximately 70% of all cutaneous melanomas. It typically presents in the fourth and fifth decades of life and is most frequently found on intermittently sun-exposed areas—commonly the lower extremities in women and the upper back in men.